Fiscal Year:	FY 2013	**Task Last Updated:**	FY 11/14/2013
PI Name:	Khusid, Boris Ph.D.		
Project Title:	Novel Electro-hydrodynamic Technology for Gas-Liquid Phase Separation in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date now shows as 8/19/2013 per NSSC (Ed., 3/21/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	James, Conrad (Sandia National Laboratories)		
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Task Description:	Due to the lack of gravity, the removal of bubbles formed in a flowing fluid presents a challenge in space applications. The most straightforward application of the proposed electro-hydrodynamic (EHD) method for bubble stripping and/or two phase flow management is in microfluidics and lab-on-a-chip (LOC) devices envisioned for space applications. Existing methods of the bubble removal systems appear to be of limited use for space applications due to different functional requirements. In the first year of our work on the project, we developed the EHD method for the field and flow driven concentration and removal of air bubbles from a droplet of aqueous-based polymer solutions, such as saliva and urine. Contrary to conventional electric field-based microgravity techniques, the main idea behind our EHD method is to apply an electric force directly to a fluid via capacitive coupling to electrodes energized by an alternating current (AC) voltage. During the second year, we developed a conveyer-belt technique that employs our EHD method for continuous processing of many droplets. The third year study focused on understanding mechanisms that underlie the EHD method. Most of experiments were conducted on suspensions of neutrally buoyant particles in order to suppress the gravity driven separation. Experiments were conducted within the range of parameters appropriate for future applications of microfluidics and LOC devices in microgravity. During the fourth year, we carried out parabolic flight tests of the EHD technology under microgravity in Houston, TX, 7/30-8/2, 2013. Flight tests demonstrated proof-of-concept for the proposed EHD control and manipulation of the fluid and bubble motion in microgravity. No electrical short circuit and sparking occurred during flight and ground operations. Flight campaign and comparative tests under normal gravity revealed the salient effects of electric, capillary and buoyancy forces and provided critical data for the development of the EHD technology for space applications. Two NJIT postdocs and a female undergraduate student were involved in the research and parabolic flight tests during the fourth year. This summer our lab hosted a team of a high-school student, a high-school teacher (minority, female) and an undergraduate student from CCNY within the New York City Research Initiative supported by the NASA Goddard Institute for Space Studies. We worked hand by hand with them on the fabrication and testing of the payload for parabolic flight tests.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The recent activity in electrically driven control, manipulation, and separation of multiphase fluid flows is motivated by myriad technologies in materials engineering, electronics, and life science. The completed study addresses the scientific and engineering issues prioritized as “Higher-Priority Areas” within the Fundamental Microgravity Research and Engineering Research Enabling Exploration programs by the Committee on Microgravity Research and as “Critical” and “Severely-Limiting” by the workshop on two-phase flow in microgravity. The research improved our understanding of the behavior of gas-liquid mixtures subject to a high-gradient electric field and facilitated the design and optimization of devices to mitigate undesirable phenomena in microgravity. Flight campaign provided critical data for the development of the EHD technology for applications in thermal management systems prioritized in the NASA Space Technology Roadmap, Technology Area 14 Thermal Management Systems. Results of flight tests indicate that the technology readiness level (TRL) has partially met the requirements for TRL-5. The unexpected behavior of 3M Novec Engineered Fluid HFE-7100 in parabolic flight tests calls for further investigation of the role of surface wettability in EHD phenomena in a bubble dispersion under microgravity. Completed parabolic flight tests in microgravity and comparative experiments under normal gravity will help design and carry out future ground and microgravity tests for technology development. The obtained results identified the effects of the electric, capillary and buoyancy forces on bubble dispersions and provided critical data for (i) modification of the setup to study the combined effects of the EHD and wetting forces on bubble dispersions in a low surface tension fluid under microgravity and (ii) testing the efficiency of EHD gas-liquid separation in microgravity. It appears that parabolic flight tests would be sufficient to reveal the effects of wetting forces on EHD phenomena in a low surface tension fluid under microgravity. Longer duration microgravity experiments aboard a suborbital reuseable launch vehicle are required for testing the efficiency of EHD gas-liquid separation. A successful completion of the proposed flight tests should advance the current knowledge of EHD phenomena in two-phase fluid systems under microgravity, demonstrate the EHD gas-liquid separation as a viable technology for thermal management systems in space applications and raise the Technology Readiness Level (TRL) of the EHD gas-liquid separation from TRL 4/5 to TRL 6. NASA's Physical Science Research Program recently awarded the PI and his team a grant NNX13AQ53G Kinetics of electric field-driven phase transitions in polarized colloids, 2013-2018, for conducting experiments on EHD of colloids in microgravity aboard the International Space Station. A mature EHD technology able to operate effectively in microgravity will benefit a broad range of spacecraft systems in long-term missions.
	The study was conducted within the scope of the originally-proposed research plan. Arnon Chait, Technical Officer, discussed opportunities at NASA GRC for applications of the proposed electro-hydrodynamic technology for gas-liquid phase separation in microgravity. The consensus was that the most straightforward application of the proposed EHD method is in lab-on-a-chip (LOC) devices envisioned for testing body fluids, such as saliva and urine, in microgravity. LOC devices inherently have air bubbles which are difficult to remove. Following this suggestion, most ground experiments were carried out on aqueous polymer solutions whose properties were similar to that of saliva. In the first year of our work on the project, we developed the EHD method for the field and flow driven concentration and removal of air bubbles from a droplet of aqueous-based solutions, such as saliva and urine, in a microgravity environment. The main idea behind our method is to apply an electric force directly to a droplet via capacitive coupling to external electrodes energized by an alternating current (AC) voltage signal. The proposed method overcomes the critical challenges facing current technologies for gas-liquid separation in microgravity and appears to meet the stringent requirements of lab-on-a-chip (LOC) devices for space applications. During the second year, we developed a conveyer-belt technique that employs the EHD method for continuous processing of many droplets. As air bubbles inevitably experience a buoyancy force in a normal gravity environment, we also performed experiments on two suspensions of neutrally buoyant particles to confirm that the EHD method does not require the presence of the buoyancy force exerted on air bubbles. Experiments demonstrated that the EHD method can operate over a broad range of electrical conductivity and viscosity of suspending fluids. Results of our experiments were discussed at the seminar given by PI in NASA's Glenn Research Center in Dec 2010. The third year study focused on understanding mechanisms that underlie the EHD method for separation of air bubbles from liquid. Most of experiments were conducted on suspensions of neutrally buoyant particles in order to suppress gas-liquid separation driven by the buoyancy force exerted on a bubble. It was found that particles would segregate only for a sufficiently

Task Progress:	strong electric field that formed a cusp-like droplet shape. Following discussions in NASA GRC, we conducted experiments on segregation and removal of air bubbles in open microchannels similar to those used by NASA GRC for testing body fluids in microgravity. The tests demonstrated that the developed EHD method can be used as an in-line debubbler in a pre-processing stage of LOC systems for bodily fluids in future space applications. However, it remained unknown (i) whether it would be possible to affect fluid flow and bubble motion in microgravity by applying voltage pulses via capacitive coupling and (ii) the extent to which the particle migration observed in a neutrally buoyant suspension subjected to an electric field under normal gravity could mimic the EHD driven behavior of a bubble dispersion in microgravity. In the fourth year, we carried out parabolic flight tests of the EHD technology under microgravity in Houston, TX, 7/30-8/2, 2013. As parabolic flight tests provide microgravity for about 13-20s per parabola, whereas the process of EHD gas-liquid phase separation takes several minutes, the objectives of flight tests were • Demonstration that the EHD method can operate in fluids with conductivity much higher than conventional electric field-based techniques; • Proving the ability of voltage pulses alternating in polarity to affect the fluid and bubble motion in microgravity; • Revealing salient effects of the buoyancy and EHD forces on bubble dispersions in microgravity and normal gravity environments. Conducting the flight tests was also supported by Space Act Agreement between NASA and NJIT for Access to a Reduced Gravity Testbed under the Flight Opportunities Program, 06/2012, PI: Boris Khusid. The flight campaign was successful. Experiments on distilled water (high surface tension fluid with electrical conductivity 5 S/m) demonstrated proof-of-concept for the EHD control of the fluid and bubble motion in microgravity by applying voltage pulses to a fluid via capacitive coupling to electrodes. While water was subjected to voltage pulses of 3-4 kV, no electrical short circuit and sparking occurred during flight and ground operations. Tests in microgravity and comparative study under normal gravity revealed the salient effects of the EHD and buoyancy forces on the fluid and bubble motion. The major trends in the EHD driven fluid and bubble motion observed in water under microgravity appeared to be consistent with expectations based on study of aqueous suspensions of neutrally buoyant particles under normal gravity and data on EHD phenomena available in the relevant literature. However, longer periods in microgravity under microgravity, up to 4.5 min aboard a suborbital vehicle, are required for quantification of the electric field effects on the flow patterns, bubble motion, and gas-liquid separation. The flight campaign also revealed the unexpected behavior of low surface tension HFE-7100, presumably caused by high wettability of the vessel wall. These observations emphasize the need to study flow phenomena in microgravity driven by the synergism of EHD and wetting forces.

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